DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (A)
Osmania University, Hyderabad-07

CLASS TEST-I

Subject : Automata Languages and Computation

Class: B.E (CSE)
Semester: IV
Academic Year: 2024-2025

Max Marks: 30
Duration: 1 hr 30 Minutes
Date : 27-03-2025 (12:00 noon to 1:30 p.m.)

Sl. No.	PART A Answer All Questions (5 x 2 = 10 Marks)	Marks	BT	CO
1.	a) Define DFA b) What is the purpose of extended transition function?	2	1 2	1 1
2.	Design a DFA to accept the language consists of even number of 0's and even number of 1's	2	2	1
3.	Consider the following ϵ - NFA Compute the ϵ - closure of each state	2	3	1
4.	Convert the regular expression **01*** to ϵ - NFA	2	3	1
5.	Verify whether the following regular expressions same or not **(01)* + (10)* + 0(10)* + 1(01)* and (ϵ+1) (01)* (ϵ+0)**	2	2	1
	PART B Answer Any Two Questions (2 x 10 = 20 M)			
6.	Theorem: If $D = (Q_D , \sum , \delta_D , \{q_0\}, F_D)$ is the DFA constructed from NFA $N = (Q_N , \sum , \delta_N , q_0 , F_N)$ by the subset construction, then prove $L(D) = L(N)$.	10	3	1
7.	Write the regular expression for the language that consists of set of all strings of 0's and 1's whose number of 0's is divisible by five.	10	2	1
8.	State and prove pumping lemma for regular languages.	10	3	1

	ϵ	a	b	c
→p	Φ	{p}	{q}	{r}
q	{p}	{q}	{r}	Φ
*r	{q}	{r}	Φ	{p}

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